# Ambient Diffusion Policy

## Imitation Learning From Suboptimal Data in Robotics

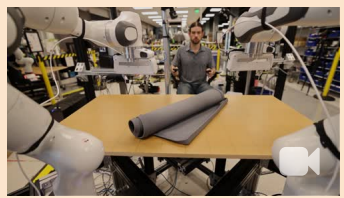
**Adam Wei**

On behalf of Nicholas Pfaff, **Thomas Cohn**, Arif Kerem Dayı, Costis Daskalakis, Giannis Daras Russ Tedrake

July 13, 2026



# "Suboptimal" Data Is Abundant

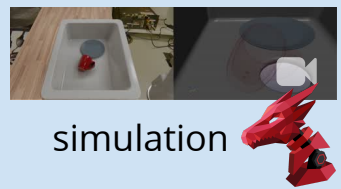


In-Distribution Data




Open-X



simulation

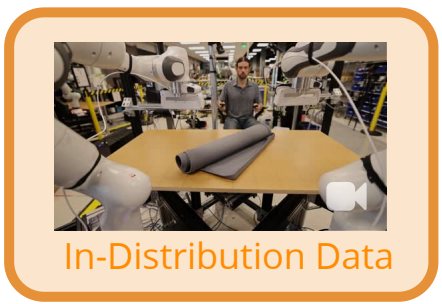
EGO-EXO4D

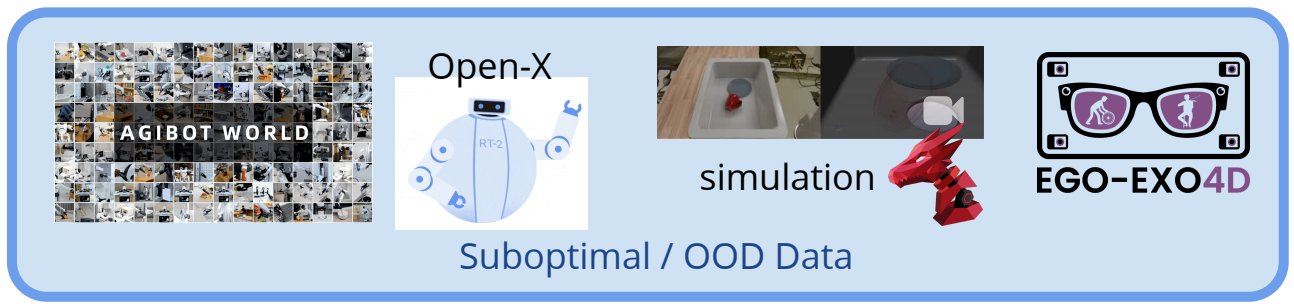
Suboptimal / OOD Data

- Noisy or non-expert teleops
- Task-level mismatch
- Camera models, poses, etc
- Changes in low-level controller
- Embodiment gap
- Sim2real gaps
- Different environment, objects, etc

**Suboptimal data:** any dataset with *out-of-distribution* actions

# Problem Statement



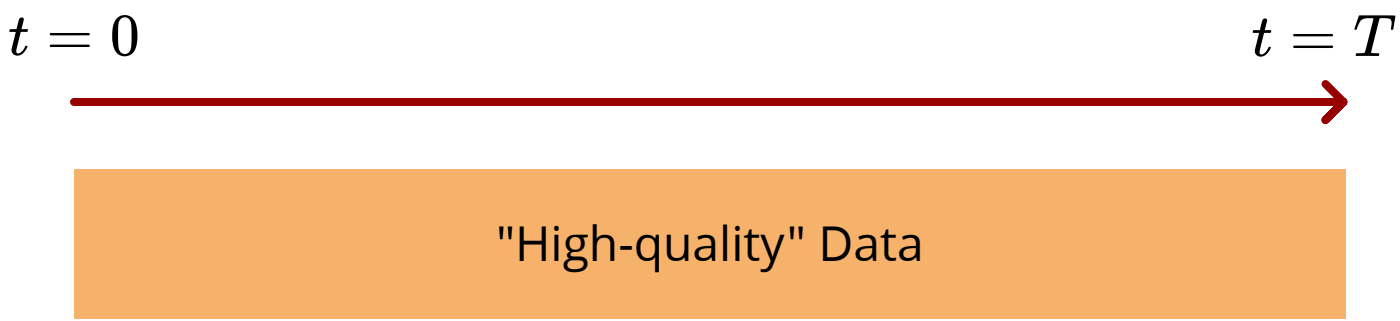




$p$ $q$

What are *principled* algorithms for learning from *suboptimal* data sources?

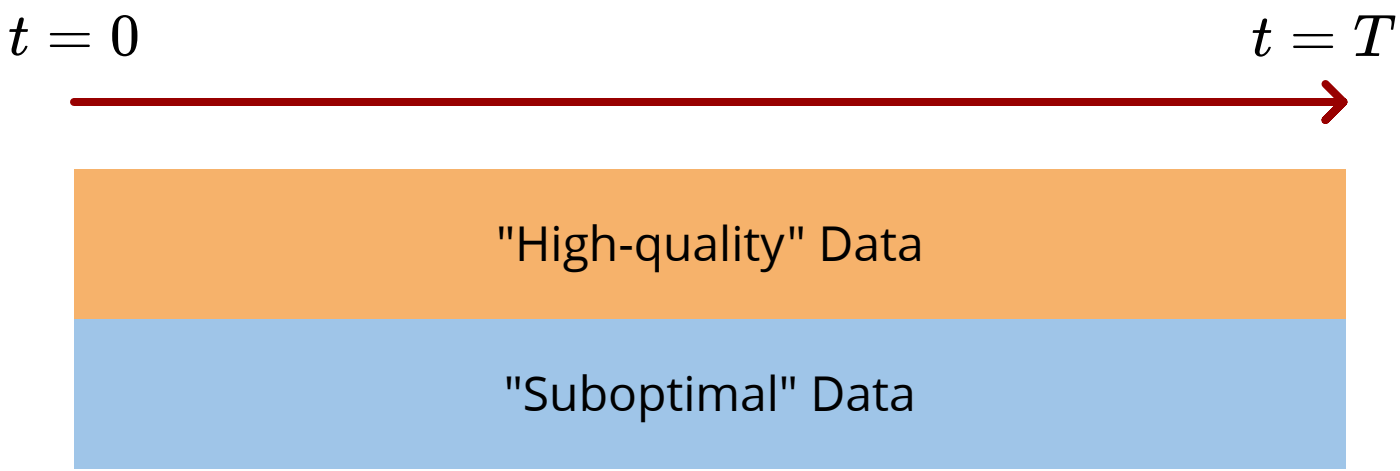
# Diffusion Training

**For all $t \in [0, T]$:** train $h_\theta(A_t, O, t) \approx \mathbb{E}[A_0 \mid A_t, O]$



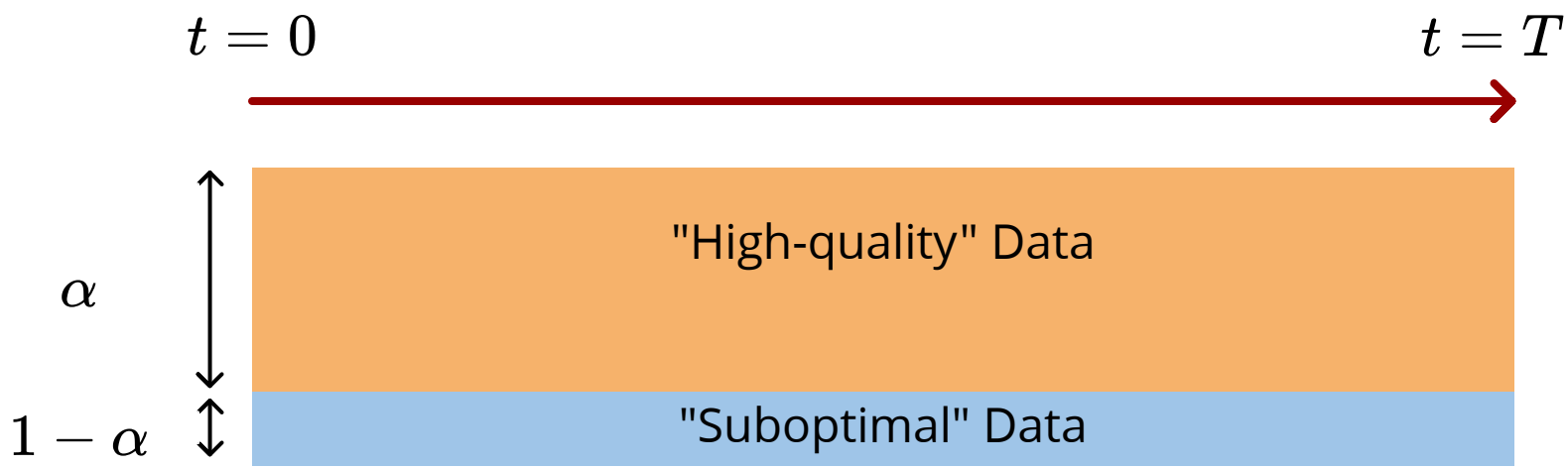
# Co-training

**For all $t \in [0, T]$:** train $h_\theta(A_t, O, t) \approx \mathbb{E}[A_0 \mid A_t, O]$



# Co-training

**For all $t \in [0, T]$:** train $h_\theta(A_t, O, t) \approx \mathbb{E}[A_0 \mid A_t, O]$




$\pi(A|O)$ learns both the good and the bad features of $q$

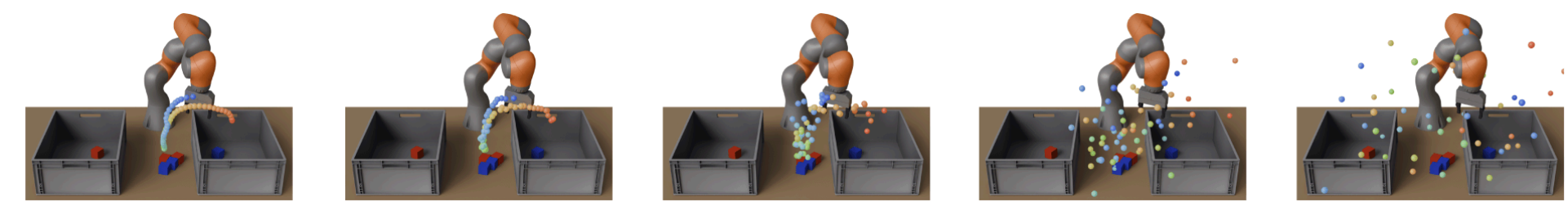
# Global-to-Local Hierachy

Action diffusion is *hierarchical*

**Low noise:**
Learn motion primitives

**High noise:**
Learn high-level planning

$t = 0$ $t = T$



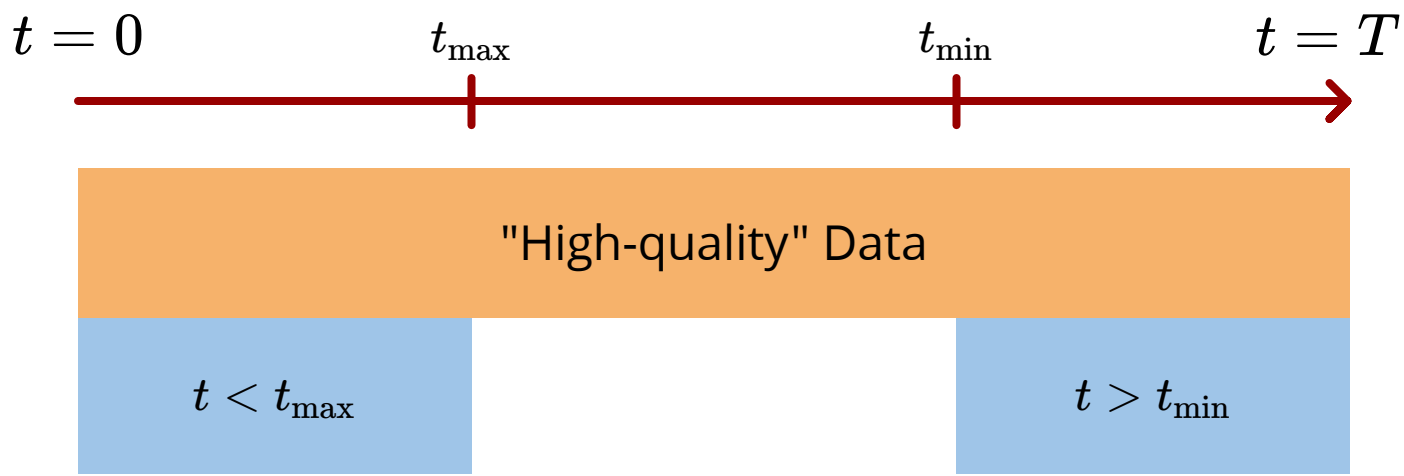
How to *grasp*? Smoothness, etc

Which block to *pick*? Which bin to *place*?

# Ambient Diffusion Policy

**Low noise:**
Learn motion primitives

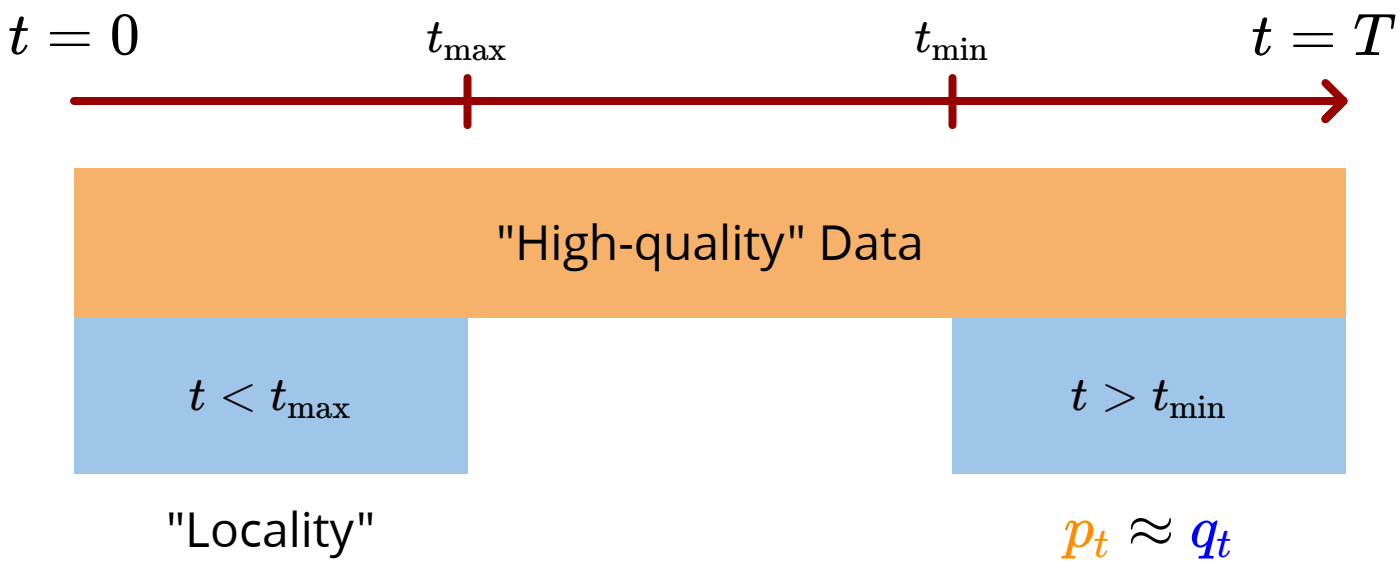
**High noise:**
Learn high-level planning




"data with high-quality *motions*,
but OOD *plans*

"data with high-quality *plans*,
but low-quality *motions*

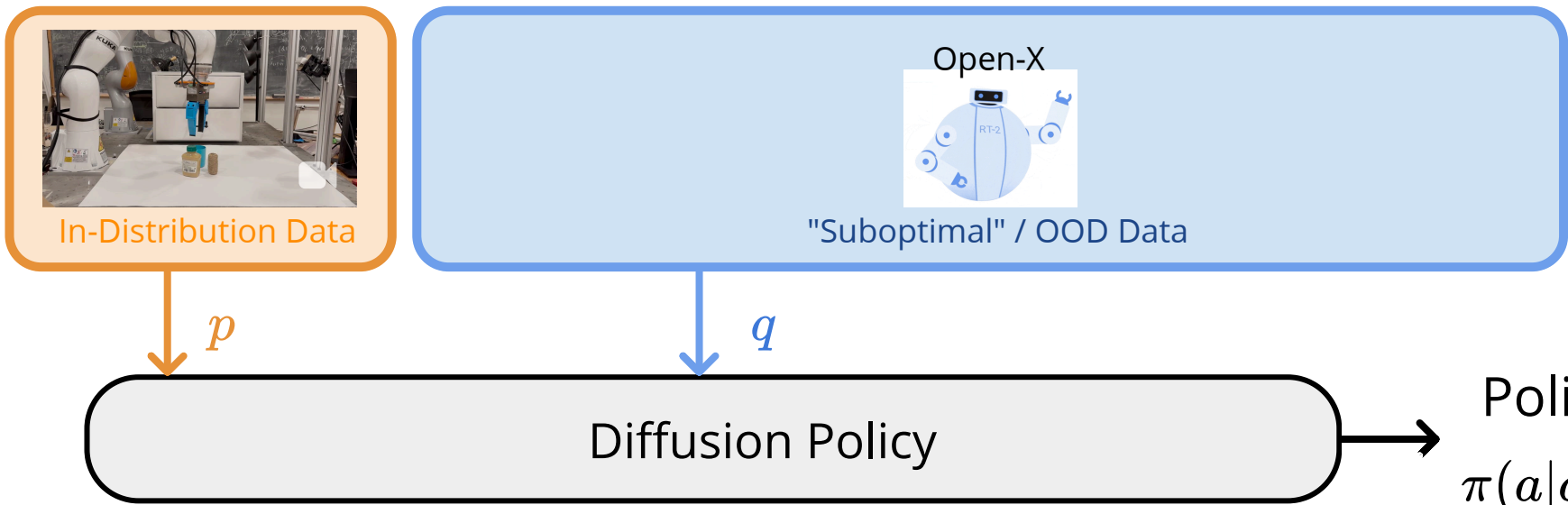
# Theoretical Interpretation




Both intervals $[0, t_{\max})$ and $(t_{\min}, T]$ have *precise* interpretations.

**Annotating $t_{\min}$ and $t_{\max}$:** automated procedure. See paper.

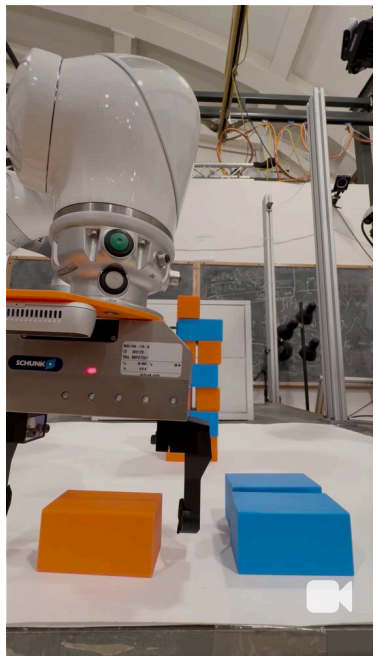
# Experiments!




- cross-embodied
- diff. teleoperators
- sim data

- mislabeled data
- diff tasks, environments, camera

# Table Cleaning & Tower Building



Tower Building

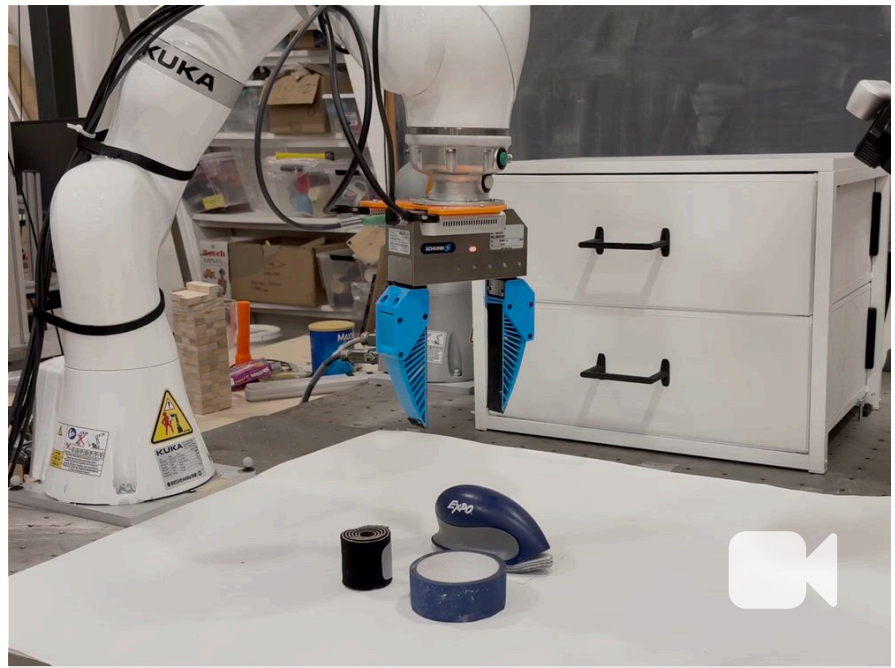
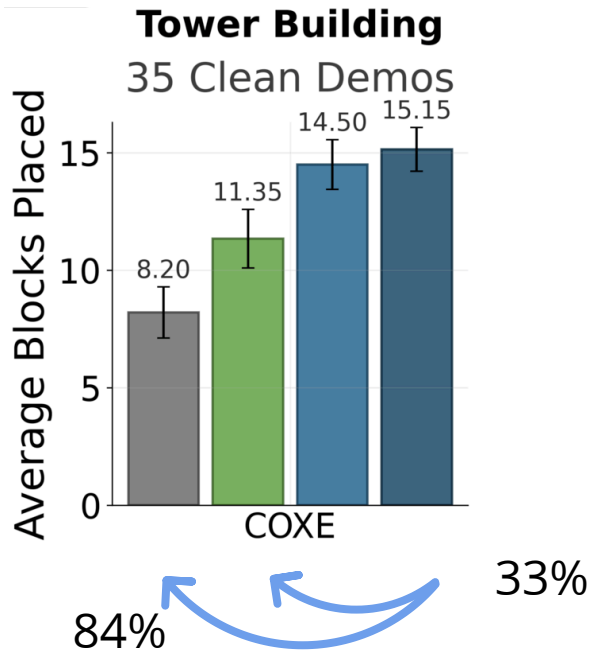


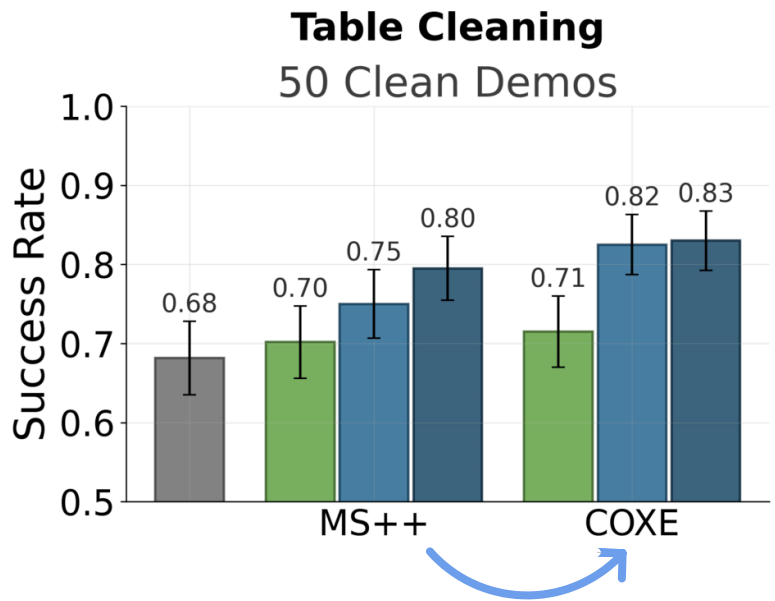

Table Cleaning

*both videos are autonomous rollouts from Ambient Diffusion Policies at 2x speed

# Ambient + Open X



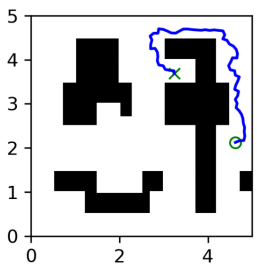
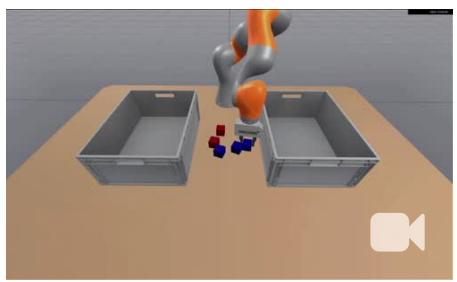




Ambient outperforms co-training by up to 33%

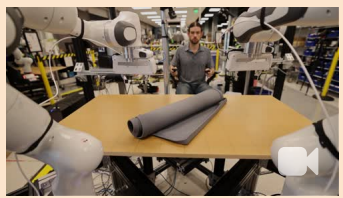
Ambient scales with *more suboptimal data*, whereas co-training plateaus

# Generality

Ambient works for *arbitrary* sources of suboptimal data

| | Noisy Teleop | | Sim2real Gap | Task Mismatch |
|---|---|---|---|---|
| |  |  |   |  |
| **Co-train:** | 99.4% Jittery | 59.9% Jittery | 84.5% | 22.7% |
| **Ambient:** | **99.5%** Smooth | **65.9%** Smooth | **93.5%** | **93.3%** |

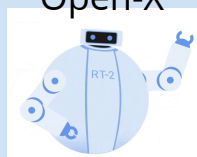
# Thank You!

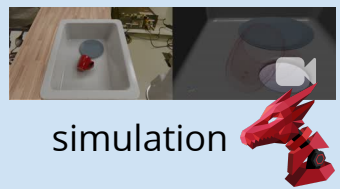


In-Distribution Data




Open-X



simulation

EGO-EXO4D

Suboptimal / OOD Data

**Ambient Diffusion Policy:**

*noise-dependent data usage* for learning from suboptimal data

**Project**